

Opinion Response-Pattern Network Analysis

DPCN Assignment 1 Report

Team: LUCH

Snehil Sanjog

Roll No: 2023102051

snehil.sanjog@students.iiit.ac.in

Manas Inamdar

Roll No: 2023102052

manas.inamdar@students.iiit.ac.in

Soham Jahagirdar

Roll No: 2023102046

soham.jahagirdar@students.iiit.ac.in

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Abstract

This report outlines the methodology, computational pipeline, and topological analysis of a 60-item opinion survey dataset. Using pairwise Spearman rank correlation and weighted Louvain community detection, we construct an opinion similarity network to identify structural bridges, core consensus points, and cross-domain thematic communities among the respondents.

1. Team Name

LUCH

2. GitHub Link to Code

https://github.com/Sylvester4582/DPCN_LUCH_A1

3. Dataset Documentation

The dataset analyzed is derived from the survey responses file `Survey_Results_UC.csv`. The instrument consists of 60 question statements divided equally across four categories: Technology (T01–T15), Education (E01–E15), Social/Ethics (S01–S15), and Environment (V01–V15).

Response labels originally included: *Strongly Disagree*, *Disagree*, *Neutral*, *Agree*, *Strongly Agree*, and *No Comments*. Missing values were preserved as `np.nan` for *No Comments* and blank cells. The ordinal mapping utilized for analysis was 1 to 5, respectively. Because the question set is ordinal rather than continuous, pairwise Spearman rank correlation was utilized as the primary similarity metric.

4. Pipeline Followed

The computational pipeline for network construction and analysis proceeded as follows:

1. **Normalization:** Survey items were mapped to short IDs (e.g., T01, E10) and Likert responses were numerically encoded.
2. **Correlation Construction:** A pairwise Spearman correlation matrix was computed across all 60 survey items.
3. **Thresholding:** The weighted adjacency matrix was defined using edges where the correlation exceeded an optimal working threshold of $T = 0.40$.
4. **Decomposition:** The largest connected component (LCC) was extracted. An isolated pair (E05, E06) and 18 additional isolates were identified.
5. **Community Detection:** Weighted Louvain community detection was applied to the LCC using seed=42, producing six distinct communities ($Q \approx 0.3676$).
6. **Topological Analysis:** Degree, strength, betweenness, closeness, eigenvector, and local clustering metrics were computed to identify structural hubs.

5. Analysis and Visualizations

5.1. Threshold Sensitivity Analysis

A threshold sensitivity analysis guided the selection of the final operating point at $T = 0.40$. As shown in Table 1, lower thresholds create dense but fragmented networks, while higher thresholds produce overly sparse graphs. The chosen threshold yields a balanced structure with a stable 40-node LCC.

Table 1: Network topology metrics across varying correlation thresholds.

Threshold	Edges	Density	Avg Degree	Components	LCC Size	Isolates	Clustering
0.30	322	0.1819	10.7333	8	53	7	0.5508
0.35	188	0.1062	6.2667	14	47	13	0.4410
0.40	91	0.0514	3.0333	20	40	18	0.2826
0.45	43	0.0243	1.4333	32	24	26	0.2558
0.50	13	0.0073	0.4333	48	7	42	0.3000
0.55	7	0.0040	0.2333	53	5	50	0.0000
0.60	2	0.0011	0.0667	58	3	57	0.0000

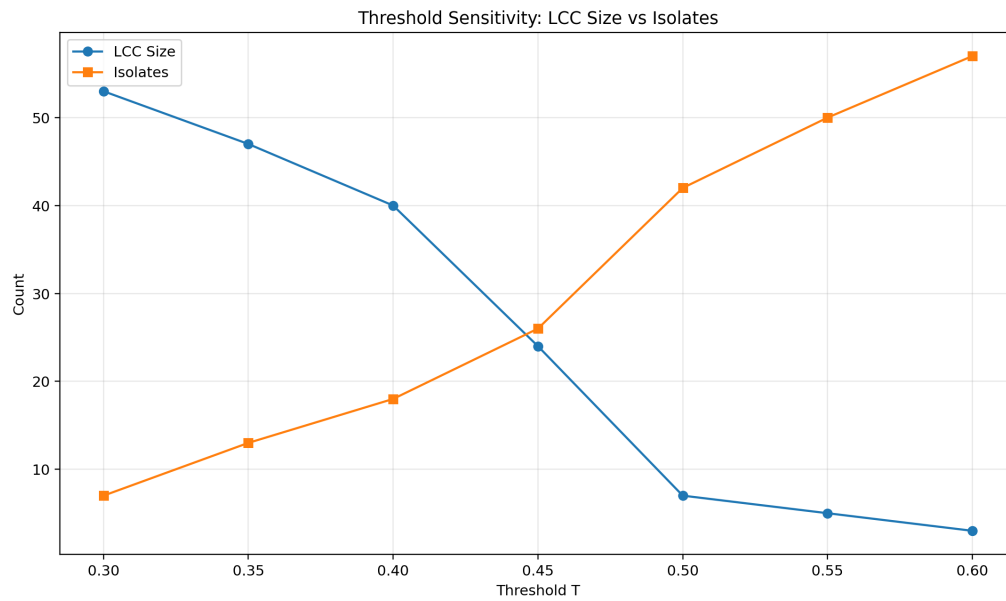


Figure 1: Threshold sweep showing the trade-off between LCC size and isolate count.

5.2. Graph Decomposition and Topology

The final graph produced an LCC of 40 nodes and 90 edges. The network is interpretable as a medium-density opinion graph: connected enough to reveal stable thematic communities, but sufficiently sparse to retain distinct local substructures.

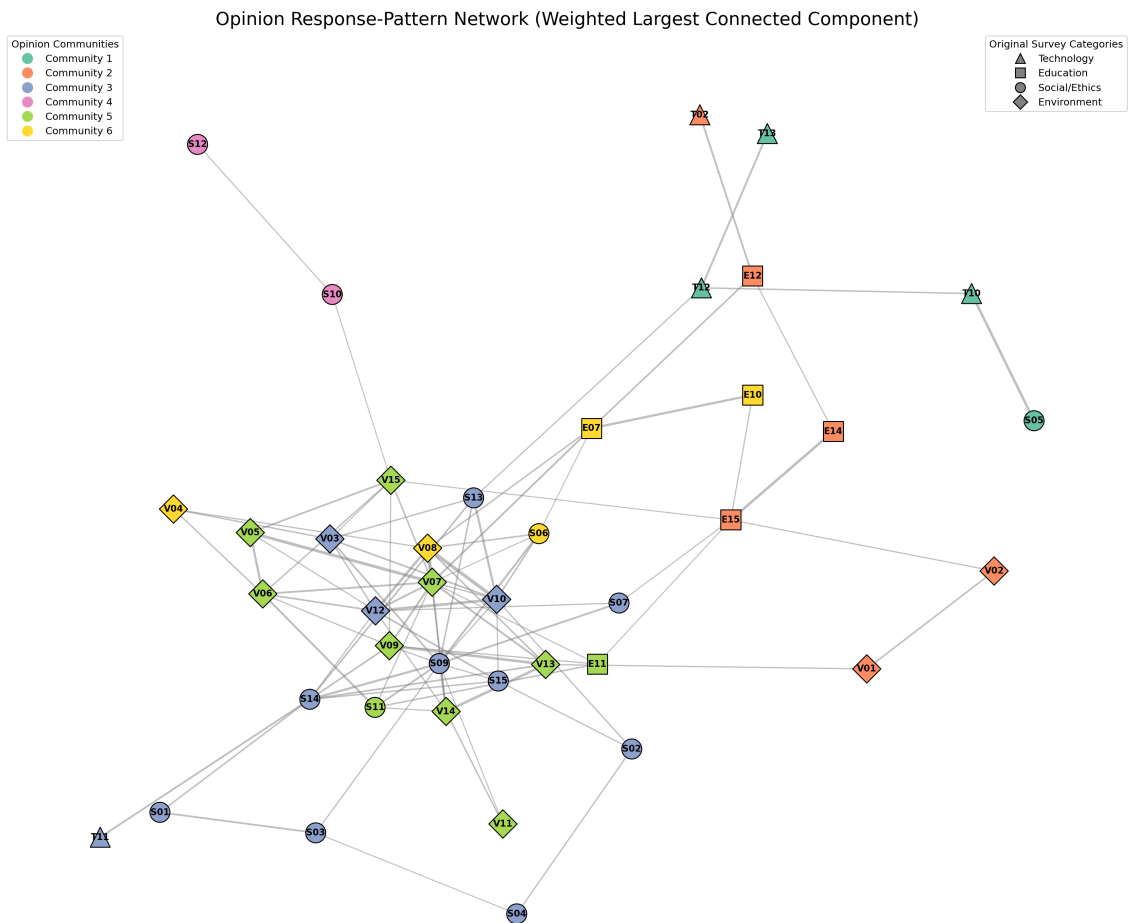


Figure 2: Weighted LCC of the opinion network at $T = 0.40$. Node shape indicates survey category; node color indicates Louvain community.

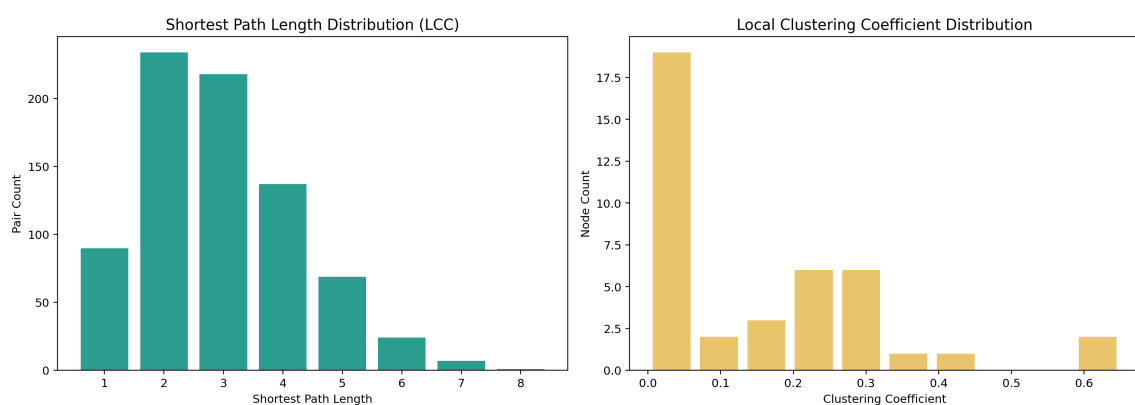


Figure 3: Distribution of shortest-path lengths and local clustering coefficients in the LCC.

5.3. Centrality Highlights

The centrality analysis identified the most important structural and consensus nodes. Node S09 acts as the primary structural bridge (highest betweenness ≈ 0.2013), connecting otherwise

separable portions of the network. Node V07 is the leading core-consensus opinion (highest eigenvector centrality ≈ 0.4041).

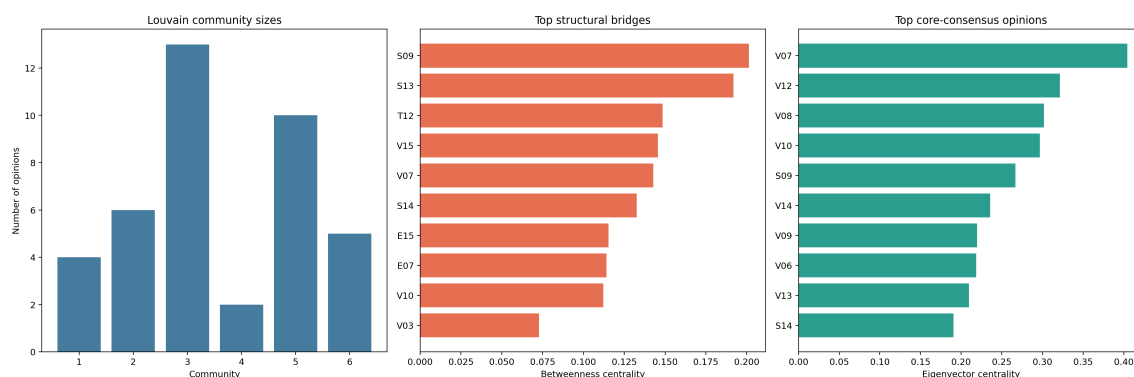


Figure 4: Community sizes, top betweenness nodes, and top eigenvector-centrality nodes.

5.4. Domain Interaction and Community Interpretation

The graph is not strictly domain-isolated. The domain mixing matrix (Figure 5) confirms robust cross-domain connectivity (e.g., between Environment and Social/Ethics), suggesting shared values across disciplines.

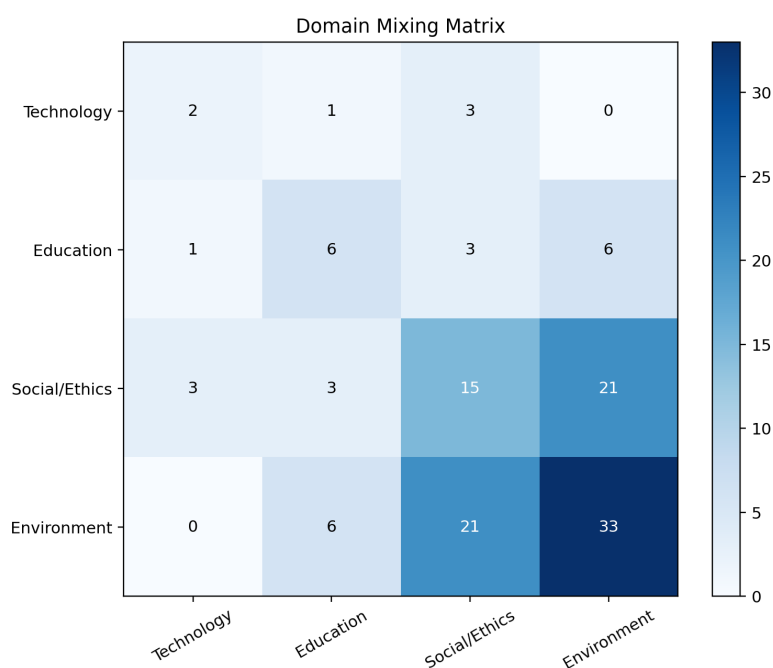


Figure 5: Domain mixing matrix highlighting cross-category edge interactions.

The 6 Louvain communities aligned with key themes:

- **Community 1:** *Data Privacy, Cybersecurity & AI Governance*
- **Community 2:** *Institutional Adaptation, AI in Education & Renewable Energy*
- **Community 3:** *Social Responsibility, Accountability & Everyday Sustainability*
- **Community 4:** *Evidence-Based Policy & Youth Civic Engagement*

- **Community 5:** *Ecological Conservation, Global Treaties & Circular Economy*
- **Community 6:** *Higher Education Innovation & Sustainable Campus Operations*

6. Results and Discussion

The results show a strong consensus around sustainability, governance, and responsibility-oriented questions. The largest and most influential community clusters revolve around social accountability, environmental stewardship, and institutional adaptation. In contrast, the isolated items point to more localized or weakly connected opinions, especially in Technology adoption debates and exam-policy questions.

The isolated pair E05 and E06 highlights a narrow curricular issue (interdisciplinary learning and required undergraduate research) that does not align strongly with the broader societal and sustainability themes dominating the LCC. Overall, the network supports the interpretation that the survey responses are not fragmented by discipline alone, but rather reflect cross-domain agreement around ethical governance, future-oriented education, and environmental responsibility.

7. Individual Contribution

- **Snehil Sanjog** (2023102051): Data preprocessing, coding, network construction, thresholding, and graph decomposition.
- **Manas Inamdar** (2023102052): Louvain community detection, topological analysis, and centrality metrics computation.
- **Soham Jahagirdar** (2023102046): Visualization, result interpretation, report drafting, and final formatting.